

NeuronBench × evolve_pysr

Meta-evolving symbolic regression for mechanistic neuron dynamics

6

mechanistically distinct
neuron worlds

4

PySR search components
co-evolved

23/25

strict held-out recoveries
in the newest run

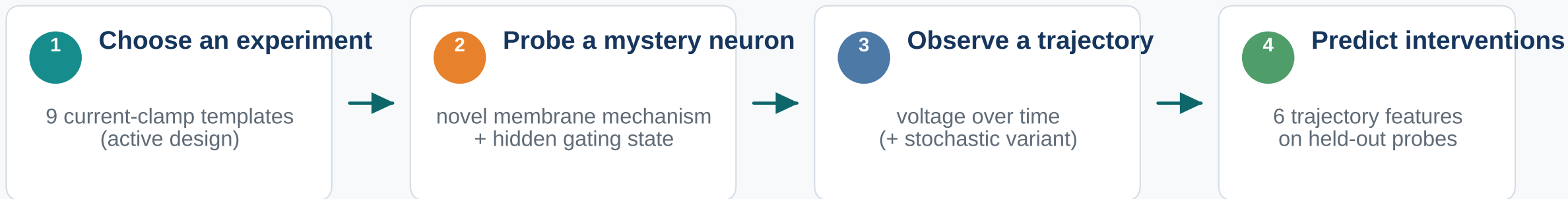
NEW: DOMAIN-BLIND

**A generic prompt still discovered the same useful search motif:
fit the residual, then add the most informative feature direction.**

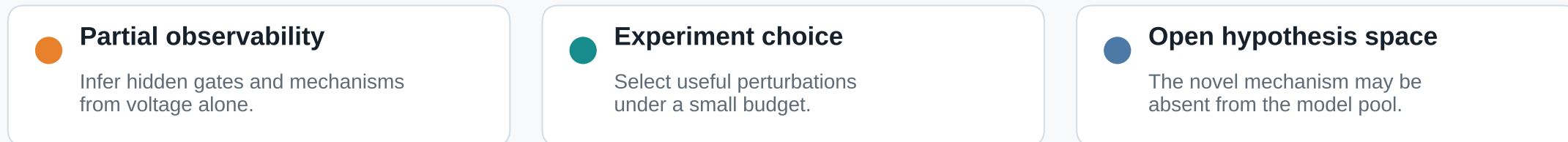
NeuronBench asks for active mechanistic discovery

ORIGINAL TASK

Six “mystery neurons” built from generalized Hodgkin–Huxley ODEs



Why it is hard



MDA combines nested SMC³ inference + LLM model proposals + Value-of-Information experiment design.

We isolate the equation-discovery problem

Original benchmark

choose current protocol

observe voltage trajectory

infer hidden gates + mechanism

forecast held-out interventions

REDUCE

Fully observable SR benchmark

sample independent neuron states

observe I_{ext} , V , and each channel-open fraction ϕ_c

fit one scalar target: dV/dt

evaluate exact physical vector field on held-out states

$$dV/dt = I_{\text{ext}} - \sum_c g_c \phi_c (V - E_c)$$

1,024

train states

16,384

held-out states

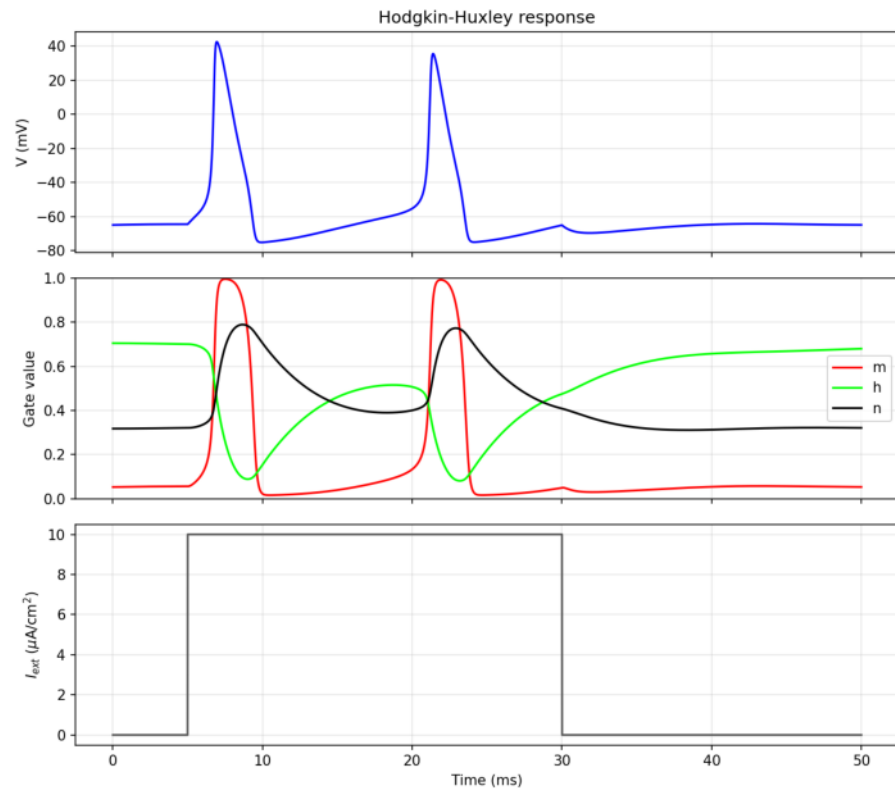
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observation noise

{+, -, ×}

operator set

Same Hodgkin–Huxley skeleton, broader discovery test



Existing project: identify HH dynamics from simulated trajectories

The shared physical law

$$C_m \frac{dV}{dt} = I_{ext} - I_{Na} - I_K - I_{leak} - I_{novel}$$

sandia_spiking

one canonical HH system
trajectory data
recover coupled ODEs

NeuronBench SR

six mechanisms
collocation states
recover membrane law

NeuronBench turns the HH demonstration into a transfer benchmark.

The benchmark is six current-balance problems

Z-rebound

1/6

$$\varphi_Z = m_Z^2 h_Z$$

$$-4 \varphi_Z (V - 120)$$

depolarization block

H-sag

2/6

$$\varphi_h = m_h$$

$$-5 \varphi_h (V + 30)$$

sag + rebound

Na-fatigue

3/6

$$\varphi_{Na} \text{ includes } s_{Na}$$

no extra current term

use-dependent rundown

Ca-rebound

4/6

$$\varphi_T = m_T^2 h_T$$

$$-3.2 \varphi_T (V - 120)$$

rebound burst

D-type

5/6

$$\varphi_D = m_D h_D$$

$$-9 \varphi_D (V + 77)$$

delayed firing

Textbook-M

6/6

$$\varphi_M = m_M$$

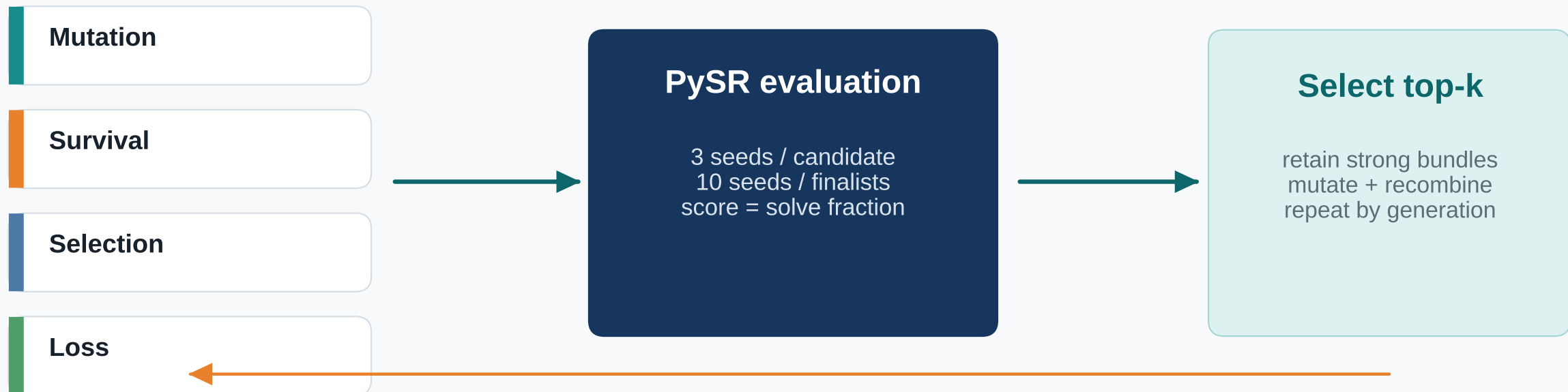
$$-2.5 \varphi_M (V + 77)$$

spike adaptation

Ground-truth complexity: 23 nodes for Na-fatigue; 31 nodes for the five worlds with an extra current.

evolve_pysr searches over the search algorithm

LLM proposes Julia operators



Transfer regimes

top-1
train on Z

top-2
train on Z + H

“top5”
train on five; hold out one

Repeated evolved motif
project residual onto a feature
then add the fitted correction

Earlier evolution transferred across unseen neurons

Manual match rate on held-out worlds



What counted as recovery?

- ① all required physical monomials
- ② no material extra monomial
- ③ clearly close coefficients
- ④ tiny numerical artifacts allowed

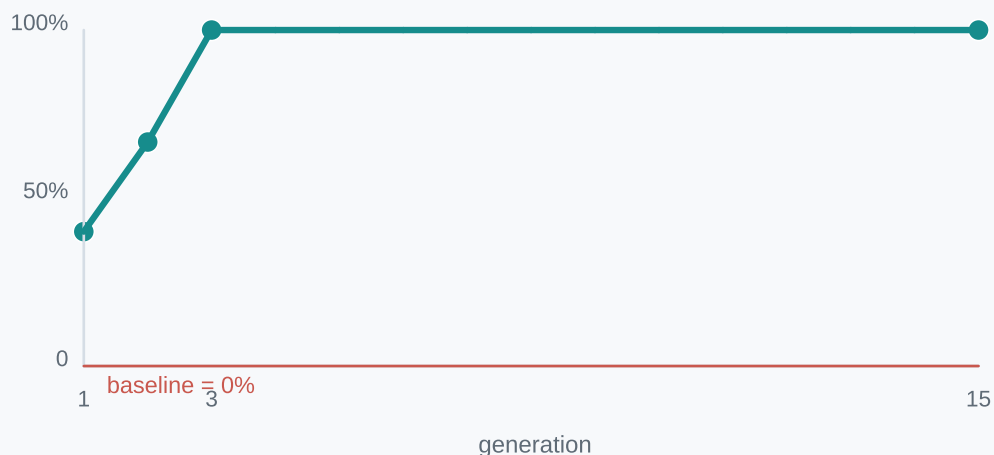
Cross-run signal: residual-guided additive mutation appears in all three best bundles.

Caveat: “top5” = one LOOCV fold

Newest result: domain-blind prompts still transfer

“Improve the algorithm’s ability to discover the expression that generated the task data.”

Best training solve rate by generation



BEST BUNDLE

residual projection mutation
+ frequency-aware selection + MSE

Held-out strict numerical recovery

H-sag



median 3.3e-10

Na-fatigue



median 2.6e-11

Ca-rebound



median 8.8e-10

D-type



median 3.3e-09

Textbook-M



median 1.3e-09

■ strict recovered ■ near-exact

23 / 25

strict recoveries
+ 2 near-exact, 0 misses

The evidence now supports a reusable search principle

- 01 Mechanistic benchmark** We preserved the exact neuron current balance while turning active, latent discovery into a controlled symbolic-regression test.
- 02 Transfer, not memorization** Algorithms evolved on 1–5 neuron worlds recover equations on unseen mechanisms; the newest run used a generic prompt.
- 03 A stable algorithmic motif** Residual-guided additive mutation repeatedly emerges because neuron currents combine as additive feature directions.

Bottom line: meta-evolution is learning a useful search heuristic—not merely receiving neuron-specific hints.